ROBOTICS ENGINEER

Abinav Pallathoor

ROS 2 · Fusion 360 · ESP32/STM32 · Python · C++

I build robots from CAD to code.

LINKEDIN ↗ | GITHUB ↗ | YOUTUBE ↗

BASE: DUBAI, UAE
CAMPUS: VELLORE, INDIA
MAIL: abinav.pallathoor@gmail.com
IN: abinav-pallathoor
GIT: AbinavPallathoor
YT: Maker portfolio
STATUS: OPEN · REMOTE

Experience & Education

■ Work □ Education

MAY 2026 - JUL 2026 ■

Research Intern
e-Yantra, IIT Bombay · eYSIP · Mumbai

> Built 3 omniwheel robots: custom PCBs, RPi 4, RPLIDAR A1, BNO055 IMU.
> Fused IMU + odometry with an EKF to fix drift; AMCL localization.
> BFS frontier exploration and a DWA local planner for narrow corridors.
> Stitched slam_toolbox maps into one global map (phase corr. + ICP).

NOV 2025 - MAY 2026 ■

Senior Product Manager
Xibotix Pvt Ltd · Tamil Nadu · Part-time

> Led the team taking rehab devices from prototype to production.
> Drove software UX and key decisions on product structure and features.

JUL 2025 - NOV 2025 ■

Software Intern
Xibotix Pvt Ltd · Tamil Nadu · On-site

> Built the Unity3D software and therapy games for Gyrosphere and ExoFist.
> ESP32 firmware, hardware integration, SQL metrics and reports.
> Designed a dynamic neck-joint mechanism in Fusion 360 for a humanoid.

JUL 2025 - AUG 2029 □

B.Tech, Computer Science
Vellore Institute of Technology · Vellore

2021 - 2025 □

High School · Computer Science
GEMS Our Own High School · Dubai

> Class XII 93.2% [100 in CS] · Class X 97% [100 in Maths]

Skills

Software · Hardware · Design

- **ROS 2** — SLAM · Gazebo
- **Unity 3D** — Games · sims
- **Python** — OpenCV · Flask
- **C / C++** — Embedded · PID
- **MCUs** — ESP32 · STM32
- **SQL** — SQLite · MySQL
- **Web** — React · Svelte
- **Fusion 360** — CAD · assembly
- **Blender** — Model · render
- **Cura** — Slicing
- **Electronics** — PCB · soldering
- **Fabrication** — 3D print · metal

Honors & Awards

12 entries · 2021 - 2026

1ST	**Equinox 2026 Hackathon**	RoboVITics × Anchor Panasonic · ₹65K
4TH	**GameDeon GameJam 2025**	Finalist · Techniche, IIT Guwahati
1ST	**4x4 in Schools Nationals 2024**	National Champions · Professional cla…
AWARD	**4x4 Innovation Award 2023**	Yas in Schools · Development class
1ST	**Novus Senior Innovation 2022**	Smart fridge that suggests recipes
1ST	**Special Olympics Unified Robotics**	Line follower · inclusive team · 2021
TOP 10	**e-Yantra Robotics Comp. '26**	IIT Bombay · of ~1,000 teams
AWARD	**RoboGenius Award 2025**	OOHS Dubai · 7 years of robotics
3RD	**GEMS Global Innovation**	GEMS Education · Jan 2024
1ST	**Inter-School STREAM Challenge**	GEMS Modern Academy · 2022-23
4TH	**World Robot Olympiad UAE 2022**	Future Innovators · healthcare robot
4TH	**FIRST LEGO League UAE Finals**	Robot game + innovation project

Projects

All work →



DER-100 — CV · IOT

OpenCV crowd detection and a live Unity sim that reroutes people via LED exit signs.



Gyrosphere — REHAB

Wrist-rehab device: built its Unity software, therapy games and SQL reports.



ExoFist — REHAB

India's first hand-rehab device: Unity games, ESP32 integration, SQL reports.



Axis X — 4X4

YAS 4x4 car with self-levelling, dual steering and an app-controlled towbar.



Axis 02 — ROBOTICS

Tower-climbing robot with a camera arm that surveys and makes quick repairs.



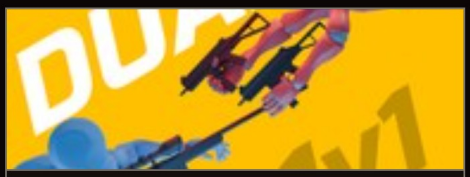
PAL — ROBOTICS

School receptionist robot that talks and guides visitors around campus.



ESP32 Watch — EMBEDDED

Ultracompact watch from common parts; case, circuit and firmware from scratch.



DUALS — GAME

1v1 local split-screen shooter built in Unity, published on Google Play.

// MORE BUILDS

CropDrop Bot — EYRC
Line-following crate delivery

Stay in the Light ↗ — GAME
GameJam finalist, on itch.io

Cable Specimen Rig — WIP
Dumbbell press, gauge marking

N20 Docking Bot — WIP
IR-beacon self-docking